

Exposure to common food additive carrageenan leads to reduced sulfatase activity and increase in sulfated glycosaminoglycans in human epithelial cells

The commonly used food additive carrageenan, including lambda (λ), kappa (κ) and iota (ι) forms, is composed of galactose disaccharides linked in alpha-1,3 and beta-1,4 glycosidic bonds with up to three sulfate groups per disaccharide residue. Carrageenan closely resembles the endogenous galactose or N-acetylgalactosamine-containing glycosaminoglycans (GAGs), chondroitin sulfate (CS), dermatan sulfate (DS), and keratan sulfate. However, these GAGs have beta-1,3 and beta-1,4 glycosidic bonds, in contrast to the unusual alpha-1,3 glycosidic bond in carrageenan. Since sulfatase activity is inhibited by sulfate, and carrageenan is so highly sulfated, we tested the effect of carrageenan exposure on sulfatase activity in human intestinal and mammary epithelial cell lines and found that carrageenan exposure significantly reduced the activity of sulfatases, including N-acetylgalactosamine-4-sulfatase, galactose-6-sulfatase, iduronate sulfatase, steroid sulfatase, arylsulfatase A, SULF-1,2, and heparan sulfamidase. Consistent with the inhibition of sulfatase activity, following exposure to carrageenan, GAG content increased significantly and showed marked differences in disaccharide composition. Specific changes in CS disaccharides included increases in di-sulfated disaccharide components of CSD (2S6S) and CS-E (4S6S), with declines in CS-A (4S) and CS-C (6S). Specific changes in heparin-heparan sulfate disaccharides included increases in 6S disaccharides, as well as increases in NS and 2S6S disaccharides. Study results suggest that carrageenan inhibition of sulfatase activity leads to re-distribution of the cellular GAG composition with increase in di-sulfated CS and with potential consequences for cell structure and function.